

Sticky Sampling

How can we compute a solution to the ϵ -AFI problem?

Sticky Sampling provides a probabilistic solution

IDEA: Probability

- ~~Precision~~ parameter $\delta \in (0, 1)$.
- Maintain frequencies for a sample of items.
- Sampling rate chosen to ensure (with probability $\geq 1 - \delta$) the frequent items are in the sample.
- 2 CASES: (1) $n = |\Sigma|$ known; (2) $n = |\Sigma|$ unknown.

Sticky Sampling: n known

Consider a stream $\Sigma = x_1, x_2, \dots, x_n$

Initialization:

- Create an empty map S (e.g., Hash Table);
- Set $t = \ln(1/(\delta\varphi))/\epsilon$.

For every item $x \in \Sigma$ do

- if $(x, f_e(x)) \in S$ then $f_e(x) = f_e(x) + 1$
else add $\{(x, f_e(x) = 1)\}$ to S with probability t/n
(start tracking x if sampled!)

approximate count of occurrences for x

Return all items x in S with $f_e(x) \geq (\varphi - \epsilon)n$

Sticky Sampling: n known

Theorem

Sticky sampling correctly solves the ϵ -AFI problem with probability at least $1 - \delta$ and requires

- $O(\log(1/(\delta\varphi))/\epsilon)$ working memory in expectation;
- 1 pass;
- $O(1)$ expected processing time per element ($S \equiv$ Hash Table).

Remark: The space is independent of n and constant for constant $\varphi, \epsilon, \delta$.

Proof

* **WORKING MEMORY** For each $x \in \Sigma$ a new entry in S is created with probability t/n
 $\Rightarrow$ The expected number of entries in S at the end of Σ is $\leq n \cdot t/n = t = \ln(1/(\delta\varphi))/\epsilon$

Proof of theorem

$X_i \equiv$ Bernoulli variable = $\begin{cases} 1 & \text{if } x_i \text{ brings a new entry in } S \\ 0 & \text{otherwise} \end{cases}$

$|S|$ at the end is $= \sum_{i=1}^n X_i$

$$E[|S|] = E\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n E[X_i] = \sum_{i=1}^n \text{Prob}(X_i=1) \leq \sum_{i=1}^n t/n = t$$

* 1 PASS : trivial

* $O(1)$ expected time per element using a Hash Table : trivial

Proof of theorem

CORRECTNESS

* Can Sticky Sampling return an item with frequency $< \varphi - \varepsilon$ (i.e., with $< (\varphi - \varepsilon)n$ occurrences)?

NO since only items with at least $(\varphi - \varepsilon)n$ occurrences are returned

$\Rightarrow$ The second requirement in the definition of ε -AFF problem is ALWAYS MET

* We only need to show that we don't miss any frequent items, with probability $\geq 1 - \delta$

Proof of theorem

$k = \# \text{ frequent items (frequency} \geq \varphi)$

Observe that $k \leq 1/\varphi$

$y = \text{an arbitrary frequent item}$

if y is not returned by the algorithm $\Rightarrow$ the first $\lceil \varepsilon n \rceil$ occurrences of y have not been sampled

$\text{Prob}(y \text{ not returned}) \leq$

$\text{Prob}(\text{The first } \lceil \varepsilon n \rceil \text{ occurrences of } y \text{ are not sampled}) =$
 $(1 - t/n)^{\lceil \varepsilon n \rceil} \leq (1 - t/n)^{\varepsilon n}$

Proof of theorem

$$\left(1 - \frac{t}{n}\right)^{\varepsilon n} = \left(1 - \frac{1}{n/t}\right)^{\frac{n}{t} \cdot \frac{t}{n} \cdot \varepsilon n} \leq \left(\frac{1}{e}\right)^{t \cdot \varepsilon}$$

We are using the INEQUALITY $(1 - 1/x)^x \leq 1/e \quad \forall x \geq 1$

$$e^{-t \cdot \varepsilon} = e^{-(\ln(1/\delta \psi)/\varepsilon) \cdot \varepsilon} = \delta \psi$$

Let $y_1, y_2, \dots, y_k$ be all frequent items ($\Rightarrow k \leq 1/\psi$)

By union bound

$$\text{Prob}(\exists y_i \text{ not returned}) \leq \sum_{i=1}^k \text{Prob}(y_i \text{ not returned}) \leq k \cdot \delta \cdot \psi \leq \delta$$

$$\Rightarrow \text{Prob}(\text{All frequent items returned}) \geq 1 - \delta \quad \square$$

Sticky Sampling: n unknown

Suppose that n is not known by the algorithm or Σ is unbounded.

GOAL: After processing $x_1, x_2, \dots, x_n$ we want to be able to derive a solution to the ϵ -AFI problem for $x_1, x_2, \dots, x_n$, for every $n \geq 1$.

IDEA: adjusted dynamically size of the stream seen so far to make sure that sampling probability is always at least t/n .

Sticky Sampling: n unknown

- Sample the $2t$ items are sampled with probability 1
- For $i = 1, 2, \dots$, sample the next batch of $2^i t$ items with probability 2^{-i}
- At the beginning of each batch **recalibrate sample S** to reflect the new sampling rate.

Specifically, **for each $(x, f_e(x)) \in S$ do:**

- Let $t_x =$ **number of tails before head** of an unbiased coin.
- **if $f_e(x) - t_x > 0$ then $f_e(x) = f_e(x) - t_x$**
else delete $(x, f_e(x))$ from S

Observations